

Fatemeh Shahrokhshahi

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RESEARCH INTERESTS

Developing computational methods to improve reasoning and reliability in large language models, with a focus on failure prediction, structured inference, and prompt-based intervention strategies.

EDUCATION

Master of Computer Engineering *Istanbul Aydin University, Istanbul, Turkey* | GPA: 4.0/4.0 Sep 2024 - Jul 2026

Bachelor of Robotic Engineering *Shahrood University of Technology, Shahrood, Iran* | GPA: 2.62/4.0 Sep 2015 - Feb 2022

RESEARCH EXPERIENCE

Can We Predict LLM Reasoning Failures? Structural Predictability of Modal Inference Errors | Apr 2026 *KI 2026: 49th German Conference on Artificial Intelligence* | Under Review

- Trained external classifiers to predict per-instance LLM failures from input-side structural features across 3,776 instances, nine models, and three prompting strategies; AUC-ROC 0.69–0.93
- Showed targeted prompting reduces failures but makes residual errors harder to predict, establishing complementarity of input- and output-side reliability mechanisms

LogiCue: Targeted Prompting for Improved Modal and Conditional Reasoning in LLMs | Dec 2025 *Procedia Computer Science by Elsevier, ACLing 2025* | Published

- Developed pattern-specific prompting framework achieving 82.8% accuracy vs. 31.9% zero-shot baseline (+50.9 pp) across seven inference patterns grounded in modal logic
- Evaluated across nine models (GPT-5, Claude Sonnet 4.5, DeepSeek Reasoner, Gemini) with statistical validation ($p < 0.01$)

Intelligent Academic Retrieval System (RAG) | Nov 2024 – Feb 2025

- Built production RAG pipeline over 512 Springer papers using LangChain and ChromaDB; embedding-based retrieval outperformed keyword matching in accuracy and latency; GitHub

Balanced K-Means for Domain Discovery in Language Models | Feb – May 2025

- Developed balanced K-means clustering preventing general-domain dominance; implemented domain transition analysis maintaining semantic coherence on C4 dataset

Earthquake Prediction Using Machine Learning (Under Preparation) | Jul 2025

- Ensemble of Random Forest, PINNs, and GNNs on AFAD data; 76.9% accuracy / 82.0% AUC with 90-day prediction horizon and population risk assessment module

Child Mental State Analysis via Drawing Behaviour | *Android Developer* 2019 – 2021

- Built Android app capturing stroke coordinates and symbol sequences; pipeline reporting structured drawing features to a clinical analysis system used by therapists

TECHNICAL SKILLS

Programming: Python, C, Java, MATLAB, SQL, FastAPI

Deep Learning Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, LangChain

Machine Learning: Representation Learning, Deep Learning, Bayesian Methods

NLP/AI Specializations: Large Language Models, Natural Language Processing, Knowledge Graph Embeddings

Research Tools: Git, Cloud Services (AWS/GCP), ChromaDB (Vector DB), Julep (Orchestration), Model Fine-tuning

Additional: Computer Vision (OpenCV), Statistical Analysis, Experimental Design

CERTIFICATIONS & SERVICE

IELTS Academic | *TRF 22IR007446SHAF010A* Oct 2022

- Overall: 7.0 (L:8, R:7, S:7, W:6) | Persian (native), English (C1), French (A2), Turkish (basic)

Academic Service |

- PC Member, EITS 2026 (Emerging Trends in Information Systems)
- Presenter, ACLing 2025, AI in Computational Linguistics, British University in Dubai. Dec 2025